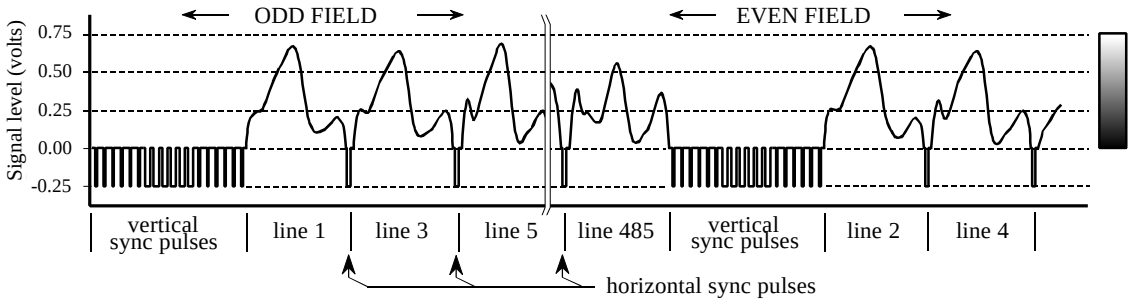

FIGURE 23-9
Composite video. The NTSC video signal consists of 30 complete frames (images) per second, with each frame containing 480 to 486 lines of video. Each frame is broken into two fields, one containing the odd lines and the other containing the even lines. Each field starts with a group of vertical sync pulses, followed by successive lines of video information separated by horizontal sync pulses. (The horizontal axis of this figure is not drawn to scale).

each frame consists of two fields, the video signal transmits 60 fields per second. Each field starts with a complex series of vertical sync pulses lasting 1.3 milliseconds. This is followed by either the even or odd lines of video. Each line lasts for 63.5 microseconds, including a 10.2 microsecond horizontal sync pulse, separating one line from the next. Within each line, the analog voltage corresponds to the grayscale of the image, with brighter values being in the direction *away* from the sync pulses. This places the sync pulses beyond the black range. In video jargon, the sync pulses are said to be *blacker than black*.

The hardware used for analog-to-digital conversion of video signals is called a **frame grabber**. This is usually in the form of an electronics card that plugs into a computer, and connects to a camera through a coaxial cable. Upon command from software, the frame grabber waits for the beginning of the next frame, as indicated by the vertical sync pulses. During the following two fields, each line of video is sampled many times, typically 512, 640 or 720 samples per line, at 8 bits per sample. These samples are stored in memory as one row of the digital image.

This way of acquiring a digital image results in an important difference between the vertical and horizontal directions. Each row in the digital image corresponds to one line in the video signal, and therefore to one row of wells in the CCD. Unfortunately, the columns are not so straightforward. In the CCD, each row contains between about 400 and 800 wells (columns), depending on the particular device used. When a row of wells is read from the CCD, the resulting line of video is filtered into a smooth analog signal, such as in Fig. 23-9. In other words, the video signal does not depend on how many columns are present in the CCD. The resolution in the horizontal direction is limited by how rapidly the analog signal is allowed to change. This is usually set at 3.2 MHz for color television, resulting in a risetime of about 100 nanoseconds, i.e., about 1/500th of the 53.2 microsecond video line.